

Eigenvalues & Eigenvectors

Introduction

Eigenvectors are the directions that remain unchanged during a transformation, even if they get longer or shorter. **Eigenvalues** are the numbers that indicate how much something stretches or shrinks during that transformation. These ideas are important in many areas of math and engineering, including studying how systems stay stable and understanding quantum physics. They're also important in fields like machine learning, where they help simplify complex data by reducing its dimensions.

Eigenvalues and Eigenvectors are the scalar and vector quantities associated with matrices used for linear transformations. The vector that only changes by a scalar factor after applying a transformation is called an eigenvector, and the scalar value attached to the eigenvector is called the eigenvalue.

*Eigenvalues are the scalar values associated with the eigenvectors in linear transformation.
The word 'Eigen' is of German Origin which means 'characteristic'.*

Introduction

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Where,

- *A is the matrix,*
- *v is associated eigenvector, and*
- *λ is scalar eigenvalue.*

Eigenvectors for square matrices are defined as non-zero vector values which when multiplied by the square matrices give the scalar multiple of the vector, i.e. we define an eigenvector for matrix A to be “v” if it specifies the condition, $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$

Eigenvector Equation

The Eigenvector equation is the equation that is used to find the eigenvector of any square matrix. The eigenvector equation is, $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$

Eigenvalues and Eigenvectors

If A is a square matrix of order $n \times n$ then we can easily find the eigenvector of the square matrix

We know that the eigenvector is given using the equation $Av = \lambda v$, for the identity matrix of order same as the order of A i.e. $n \times n$ we use the following equation,

$$(A - \lambda I)v = 0$$

Solving the above equation we get various values of λ as $\lambda_1, \lambda_2, \dots, \lambda_n$ these values are called the eigenvalues and we get individual eigenvectors related to each eigenvalue.

Simplifying the above equation we get v which is a column matrix of order $n \times 1$ and v is written as,

$$v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ \vdots \\ v_n \end{bmatrix}$$

The diagram illustrates the eigenvalue equation $Ax = \lambda x$. On the left, a green **A** is labeled "n × n Matrix" with a green line. Next to it is a red **x** with a red arrow above it, labeled "Eigenvector" with a red line. This is followed by an equals sign. On the right, a blue **λ** is labeled "Eigenvalue" with a blue line. Next to it is another red **x** with a red arrow above it, labeled "Eigenvector" with a red line.

$$\begin{array}{c} \text{A} \\ \hline n \times n \\ \text{Matrix} \end{array} \begin{array}{c} \xrightarrow{\hspace{1cm}} \\ \text{x} \\ \hline \text{Eigenvector} \end{array} = \begin{array}{c} \lambda \\ \hline \text{Eigenvalue} \end{array} \begin{array}{c} \xrightarrow{\hspace{1cm}} \\ \text{x} \\ \hline \text{Eigenvector} \end{array}$$

Eigenvalues & Eigenvectors

Notes:

- The German prefix “eigen” roughly translates to “self” or “own”.
- Eigenvalues and eigenvectors are only for square matrices.
- Eigenvectors are *by definition nonzero*. Eigenvalues may be equal to zero.
- We do not consider the zero vector to be an eigenvector: since $A0 = 0 = \lambda 0$ for *every* scalar λ , the associated eigenvalue would be undefined.

Eigenvalues & Eigenvectors

Recall the transformation involving multiplication of a vector and a matrix $\Rightarrow \mathbf{x} \mapsto A\mathbf{x}$

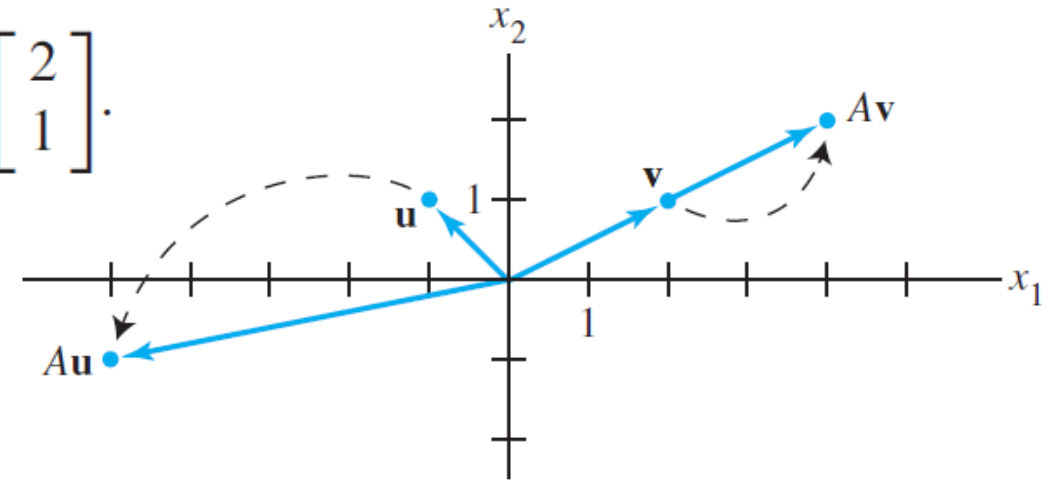
Although a transformation $\mathbf{x} \mapsto A\mathbf{x}$ may move vectors in a variety of directions, it often happens that there are special vectors on which the action of A is quite simple.

Example:

$$\text{Let } A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \text{ and } \mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Comparing the images of $\mathbf{u}$ and $\mathbf{v}$, A only stretches,
Or dilates, $\mathbf{v}$.

$$A\mathbf{v} = 2\mathbf{v}$$



(Transformation of vectors by a matrix into scalar multiple of themselves) Effects of multiplication by A .

Eigenvalues & Eigenvectors

EXAMPLE · Consider the matrix

$$A = \begin{pmatrix} 2 & 2 \\ -4 & 8 \end{pmatrix} \quad \text{and vectors} \quad v = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad w = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Which are eigenvectors? What are their eigenvalues?

SOLUTION

We have

$$Av = \begin{pmatrix} 2 & 2 \\ -4 & 8 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4v.$$

Hence, v is an eigenvector of A , with eigenvalue $\lambda = 4$.

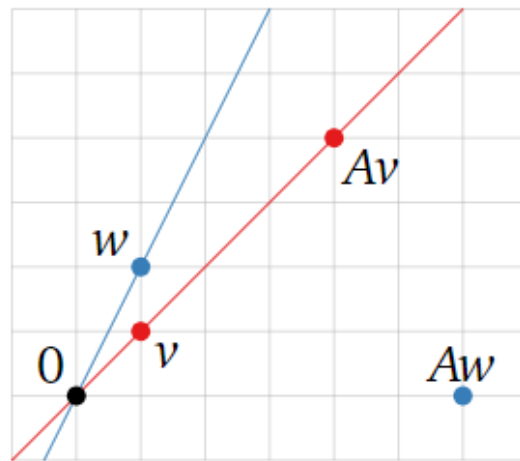
Eigenvalues & Eigenvectors

SOLUTION (Continued...)

On the other hand,

$$Aw = \begin{pmatrix} 2 & 2 \\ -4 & 8 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}.$$

which is not a scalar multiple of w . Hence, w is not an eigenvector of A .



v is an eigenvector

w is not an eigenvector

Eigenvalues & Eigenvectors

EXAMPLE 2 Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$. Are $\mathbf{u}$ and $\mathbf{v}$ eigenvectors of A ?

SOLUTION

$$A\mathbf{u} = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix} = -4\mathbf{u}$$

$$A\mathbf{v} = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -9 \\ 11 \end{bmatrix} \neq \lambda \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

Thus $\mathbf{u}$ is an eigenvector corresponding to an eigenvalue (-4) , but $\mathbf{v}$ is not an eigenvector of A , because $A\mathbf{v}$ is not a multiple of $\mathbf{v}$. ■

Eigenvalues & Eigenvectors

EXAMPLE : Consider the matrix

$$A = \begin{pmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{pmatrix} \quad \text{and vectors} \quad v = \begin{pmatrix} 16 \\ 4 \\ 1 \end{pmatrix} \quad w = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}.$$

Which are eigenvectors? What are their eigenvalues?

SOLUTION We have

$$Av = \begin{pmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 16 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 32 \\ 8 \\ 2 \end{pmatrix} = 2v.$$

Hence, v is an eigenvector of A , with eigenvalue $\lambda = 2$.

Eigenvalues & Eigenvectors

SOLUTION (Continued...)

On the other hand,

$$Aw = \begin{pmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 28 \\ 1 \\ 1 \end{pmatrix},$$

which is not a scalar multiple of w . Hence, w is not an eigenvector of A .

Eigenvalues & Eigenvectors

EXAMPLE , Let

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix} \quad v = \begin{pmatrix} -3 \\ 1 \end{pmatrix}.$$

Is v an eigenvector of A ? If so, what is its eigenvalue?

SOLUTION The product is

$$Av = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix} \begin{pmatrix} -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0v.$$

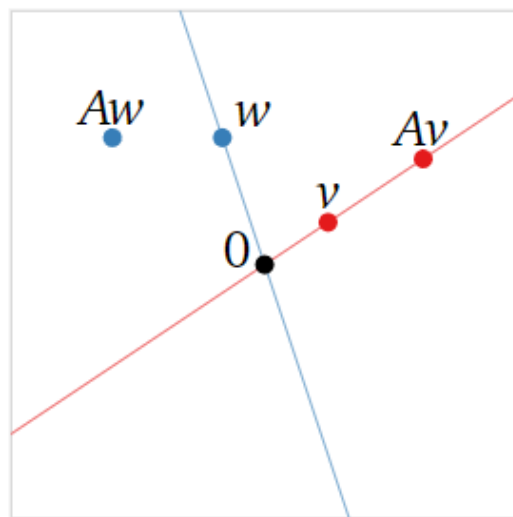
Hence, v is an eigenvector with eigenvalue *zero*.

As noted above, an *eigenvalue* is allowed to be zero, but an *eigenvector* is not.

Eigenvalues & Eigenvectors

Note:

To say that $Av = \lambda v$ means that Av and λv are *collinear with the origin*. So, an eigenvector of A is a nonzero vector v such that Av and v lie on the same line through the origin. In this case, Av is a scalar multiple of v ; the eigenvalue is the scaling factor.



v is an eigenvector

w is not an eigenvector

Graphical Examples

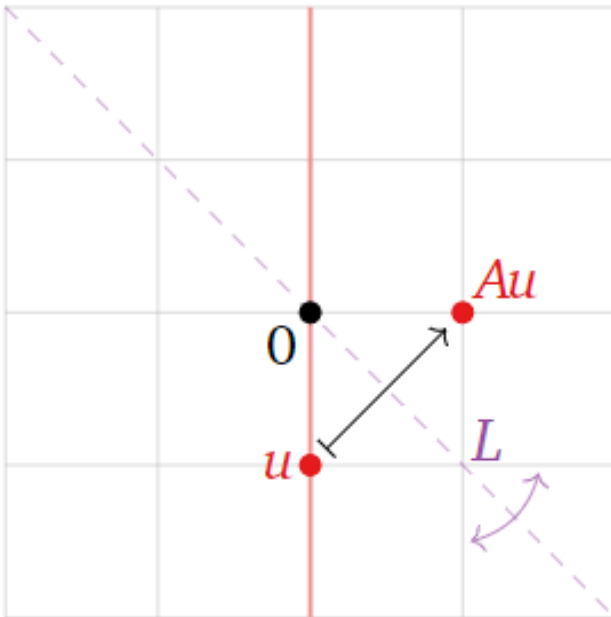
Graphical Examples for Various Transformations

1. Reflection
2. Projection
3. Rotation
4. Dilation
5. Shear

Graphical Examples

1. Reflection

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that reflects over the line L defined by $y = -x$, and let A be the matrix for T .



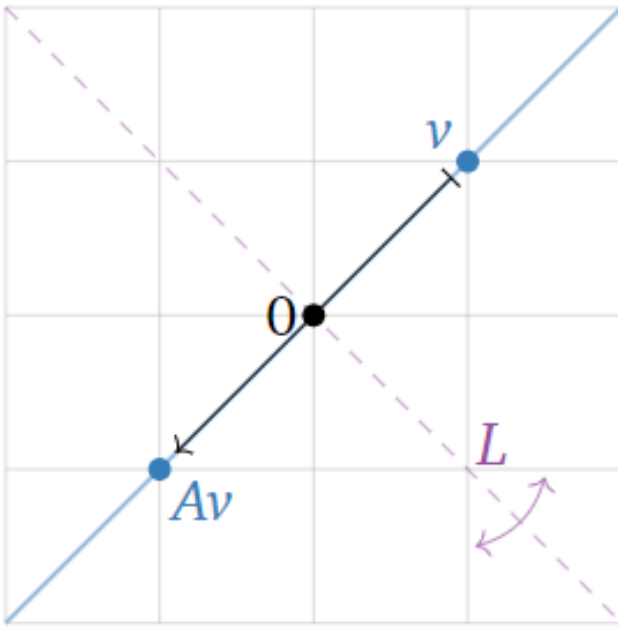
Q1: Is vector u an eigenvector?

The vector u is not an eigenvector, because Au is not collinear with u and the origin.

Graphical Examples

1. Reflection (Continued...)

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that reflects over the line L defined by $y = -x$, and let A be the matrix for T .



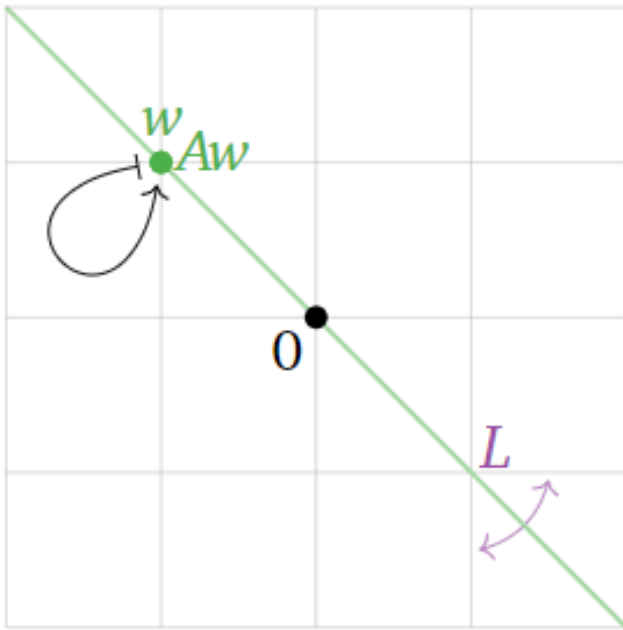
Q2: Is vector v an eigenvector?

The vector v is an eigenvector because Av is collinear with v and the origin. The vector Av has the same length as v , but the opposite direction, so the associated eigenvalue is -1 .

Graphical Examples

1. Reflection (Continued...)

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that reflects over the line L defined by $y = -x$, and let A be the matrix for T .



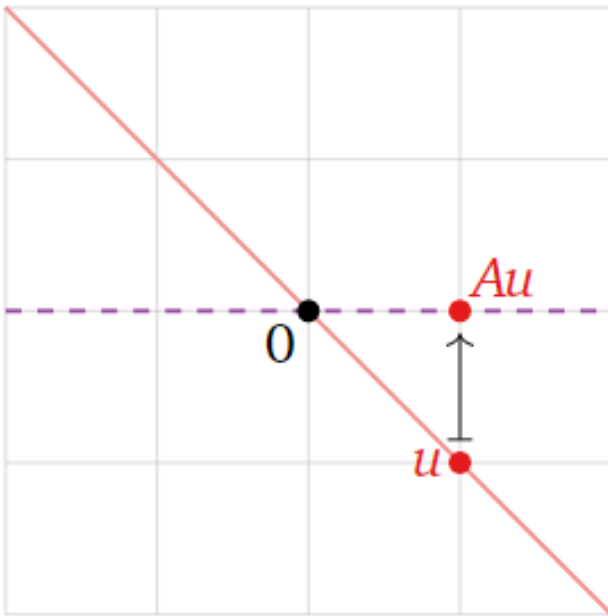
Q3: Is vector w an eigenvector?

The vector w is an eigenvector because Aw is collinear with w and the origin: indeed, Aw is equal to w ! This means that w is an eigenvector with eigenvalue 1.

Graphical Examples

2. Projection

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that projects a vector vertically onto the x-axis, and let A be the matrix for T .



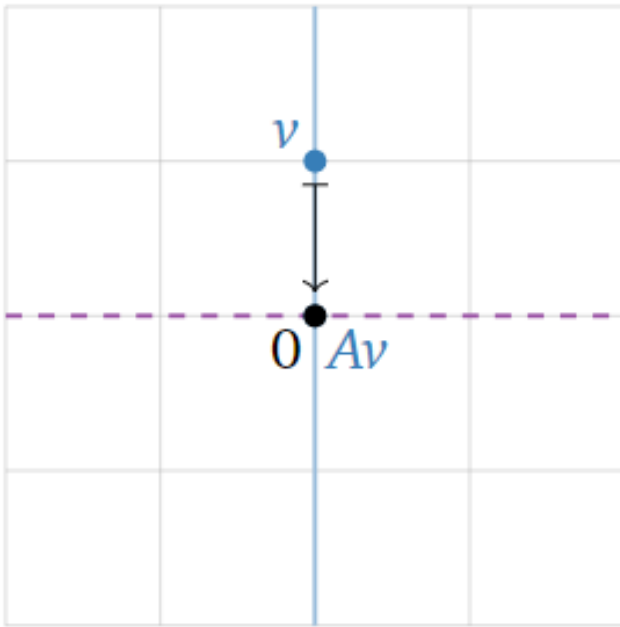
Q1: Is vector u an eigenvector?

The vector u is not an eigenvector, because Au is not collinear with u and the origin.

Graphical Examples

2. Projection (Continued...)

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that projects a vector vertically onto the x -axis, and let A be the matrix for T .



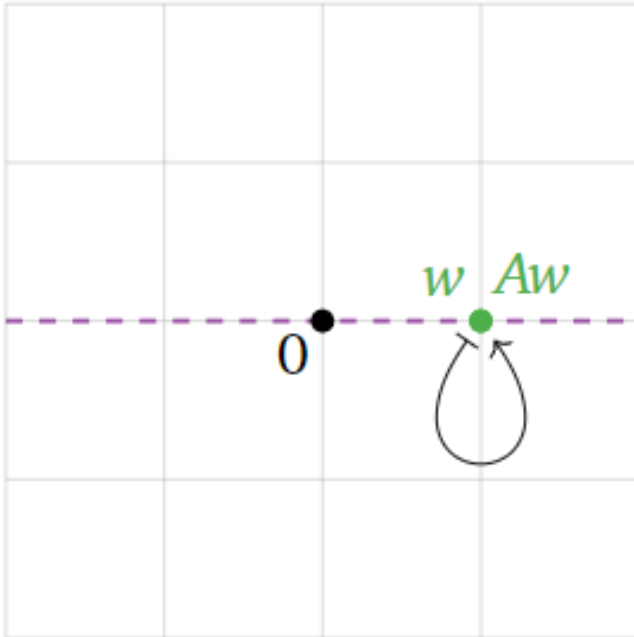
Q2: Is vector v an eigenvector?

The vector v is an eigenvector. Indeed, Av is the zero vector, which is collinear with v and the origin; since $Av = 0v$, the associated eigenvalue is 0.

Graphical Examples

2. Projection (Continued...)

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that projects a vector vertically onto the x -axis, and let A be the matrix for T .



Q3: Is vector w an eigenvector?

The vector w is an eigenvector because Aw is collinear with w and the origin: indeed, Aw is equal to w ! This means that w is an eigenvector with eigenvalue 1.

Graphical Examples

Notes:

For a Reflection Transformation

It appears that all eigenvectors lie either on L , or on the line perpendicular to L . The vectors on L have eigenvalue 1, and the vectors perpendicular to L have eigenvalue -1 .

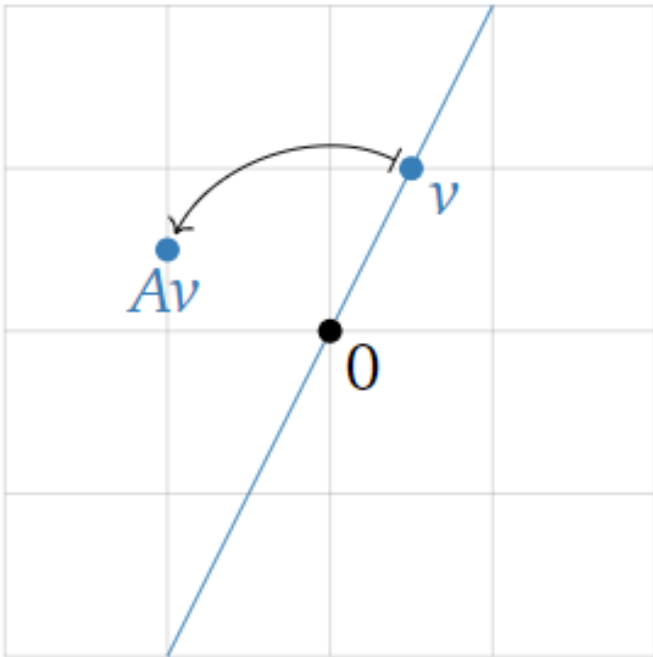
For a Projection Transformation

It appears that all eigenvectors lie on the x -axis or the y -axis. The vectors on the x -axis have eigenvalue 1, and the vectors on the y -axis have eigenvalue 0.

Graphical Examples

3. Rotation

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that rotates counterclockwise by 90° , and let A be the matrix for T .



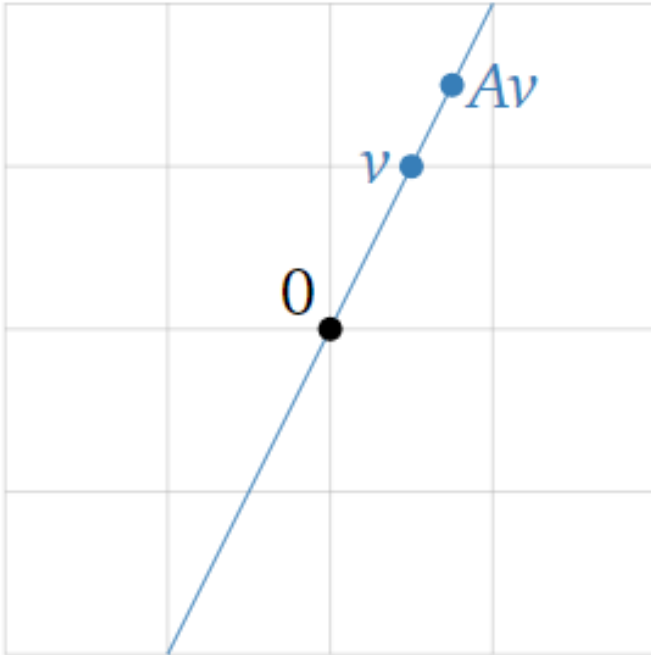
Q: Is vector v an eigenvector?

If v is any nonzero vector, then Av is rotated by an angle of 90° from v .
Therefore, Av is not on the same line as v , so v is not an eigenvector.

Graphical Examples

4. Dilation

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that dilates by a factor of 1.5, and let A be the matrix for T .



Q: Is vector v an eigenvector?

We have

$$Av = T(v) = 1.5v$$

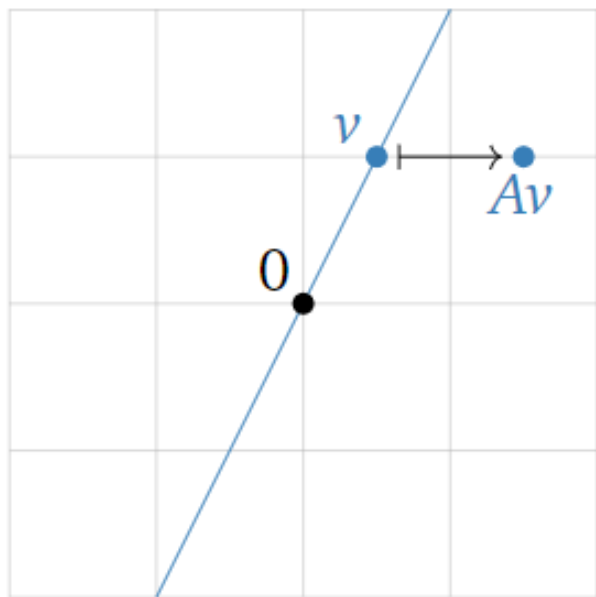
for every vector v in $\mathbb{R}^2$.

Therefore, by definition every nonzero vector is an eigenvector with eigenvalue 1.5.

Graphical Examples

5. Shear

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and let $T(x) = Ax$, so T is a shear in the x -direction.



In equations, we have

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ y \end{pmatrix}.$$

This tells us that a shear takes a vector and adds its y -coordinate to its x -coordinate. Since the x -coordinate changes but not the y -coordinate, this tells us that any vector v with nonzero y -coordinate cannot be collinear with Av and the origin.

On the other hand, any vector v on the x -axis has zero y -coordinate, so it is not moved by A . Hence v is an eigenvector with eigenvalue 1.

Graphical Examples

Notes:

- The identity matrix has the property that $I_n v = v$ for all vectors v in $\mathbf{R}^n$. We can write this as $I_n v = 1 \cdot v$, so *every* nonzero vector is an eigenvector with eigenvalue 1.

For an Identity Matrix

Every nonzero vector is an eigenvector of the identity matrix.

For a Dilation Transformation

Every nonzero vector is an eigenvector of a dilation matrix.